

Fitting logistic regression models

Last time: Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 3$ and $\beta_1 = -0.5$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7	
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}}$	0.818	0.731	0.622	0.5	0.378

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.818)^3 (0.731)^2 (0.622)^3 (1 - 0.622)^2 (0.5) (1 - 0.5) \\ \cdot (0.378) (1 - 0.378)^2$$

$$= 0.000368$$

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 4$ and $\beta_1 = -0.75$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3		4	5	6	7
π	$\frac{\exp\{4 - 0.75(3)\}}{1 + \exp\{4 - 0.75(3)\}}$		$= 0.852$	0.731	0.562	0.378
			0.223			

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.852)^3 (0.731)^2 (0.562)^3 (1 - 0.562)^2 (0.378) (1 - 0.378) \\ \cdot (0.223) (1 - 0.223)^2$$

$$= 0.000356$$

Maximum likelihood for logistic regression

Likelihood:

- + For estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, $\hat{\pi} = \frac{\exp\{\hat{\beta}_0 + \hat{\beta}_1 x\}}{1 + \exp\{\hat{\beta}_0 + \hat{\beta}_1 x\}}$
- + $L(\hat{\beta}_0, \hat{\beta}_1) = P(\text{data})$

Maximize:

- + Choose $\hat{\beta}_0, \hat{\beta}_1$ to maximize $L(\hat{\beta}_0, \hat{\beta}_1)$

In the class activity examples, we have only considered a few different possibilities for $\hat{\beta}_0$ and $\hat{\beta}_1$. In reality, we consider all real numbers as possible values.

Fitting logistic regression

Logistic regression with maximum likelihood

- + **Model:**

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 x$$

- + **Measure of fit:**

$L(\hat{\beta}_0, \hat{\beta}_1)$ = probability of observed data for estimates $\hat{\beta}_0, \hat{\beta}_1$

- + **Optimize:** choose $\hat{\beta}_0, \hat{\beta}_1$ to maximize $L(\hat{\beta}_0, \hat{\beta}_1)$

Linear regression with least squares

- + **Model:** $y = \beta_0 + \beta_1 x + \varepsilon$

- + **Measure of fit:**

$$SSE = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- + **Optimize:** choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize SSE

Regression assumptions

What assumptions did we make for linear regression models?

Regression assumptions

What assumptions did we make for linear regression models?

- + Shape
- + Constant variance
- + Normality
- + Independence
- + Randomness

Not all these assumptions translate to logistic regression.

Assumptions: logistic vs. linear regression

Logistic regression:

$$\pi = P(y = 1)$$

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 x$$

- + shape
- + randomness
- + independence

Linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

$$\varepsilon \sim N(0, \sigma_\varepsilon^2)$$

- + shape
- + randomness
- + independence
- + constant variance
- + normality

The constant variance and normality assumptions apply specifically to the noise term ε

Data

Relationship between the length of a putt, and whether it was made:

- + x = length (feet)
- + y = result (success = 1, failure = 0)

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 x$$

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134

Assessing the shape assumption

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With scatterplots and residual plots.

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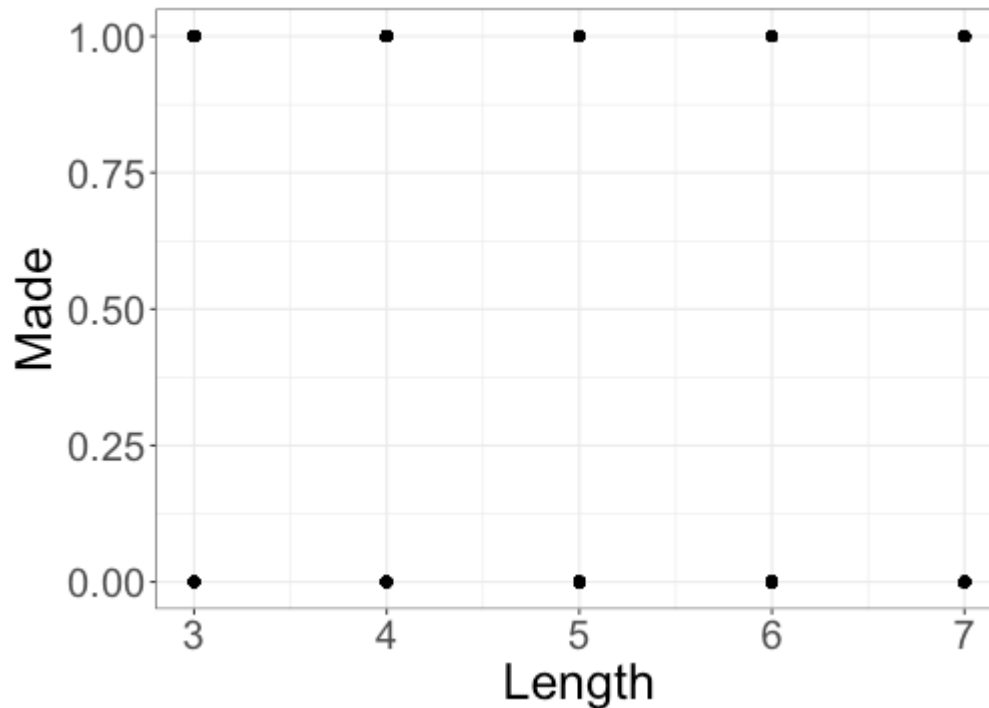
Can we use these diagnostics for logistic regression?

No:

- + the response is binary
- + the shape assumption is that the *log odds* (not the response) are a linear function of the predictor(s)

Assessing the shape assumption

Scatterplots don't work for logistic regression:



Shape

- + Shape assumption is that the log-odds are (at least approximately) a linear function of x : $\log(odds) \approx \beta_0 + \beta_1 x$
- + We can assess the shape assumption by plotting the **empirical log-odds** (aka empirical logits)
- + Example: relationship between the length of a putt and whether it was made

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Empirical log-odds

Step 1: estimate the probability of success for each length of putt

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{p}$	0.832	0.739	0.565	0.488	0.328

Empirical log-odds

Step 2: convert empirical probabilities to empirical odds

Length	3	4	5	6	7
Number of successes	84	88	61	61	44
Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{p}$	0.832	0.739	0.565	0.488	0.328
Odds $\frac{\hat{p}}{1 - \hat{p}}$	4.941	2.839	1.298	0.953	0.489

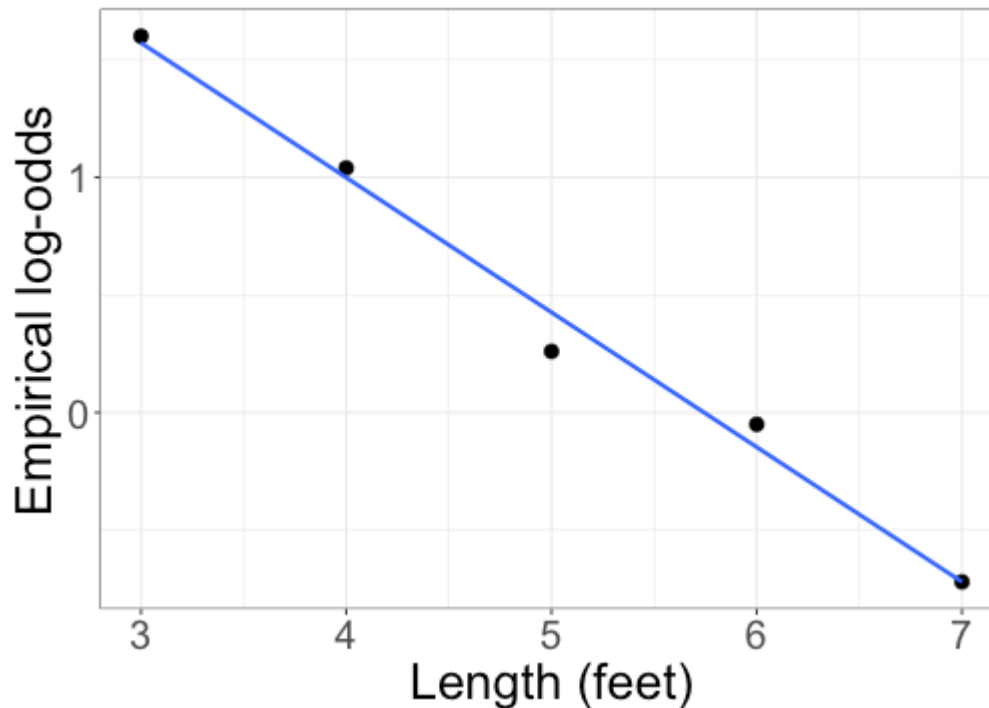
Empirical log-odds

Step 3: calculate empirical log-odds

Length	3	4	5	6	7
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Number of failures	17	31	47	64	90
Total	101	119	108	125	134
Probability of success $\hat{p}$	0.832	0.739	0.565	0.488	0.328
Odds $\frac{\hat{p}}{1 - \hat{p}}$	4.941	2.839	1.298	0.953	0.489
Log-odds $\log\left(\frac{\hat{p}}{1 - \hat{p}}\right)$	1.60	1.04	0.26	-0.05	-0.72

Empirical log-odds

Step 4: plot empirical log-odds against predictor



The shape assumption is reasonable if the empirical log-odds plot looks roughly linear.

Empirical log odds plots

Now let's try this with the med school admissions data:

GPA	2.26	2.42	2.48	2.52	2.55	2.56	...
Admit = 0	1	1	1	1	1	1	...
Admit = 1	0	1	0	0	0	0	...

What problem do I run into?

Empirical log odds plots

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Admit = 0	1	1	1	1	1	1	...
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What problem do I run into?

There are a small number of observations for each GPA, so we can't calculate empirical log odds for each GPA.

Solution: Divide the data into *bins*

Empirical log odds plots

- + Divide observations into bins of equal size, based on values of predictor X
- + Calculate empirical log odds in each bin
- + Plot empirical log odds

